BayesINE4. Modelos de regresión lineal

Intro a Estadística Bayesiana

DataLab

Abril 2026

ICMAT-CSIC

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Chapter objectives

Objectives

- Revisit main concepts in Bayesian inference through a simple but very useful model: Linear regression
  - Prediction
- Hints on hierarchical models
- Hints on empirical Bayes
- A lab reinforces concepts operationally

Basic setup

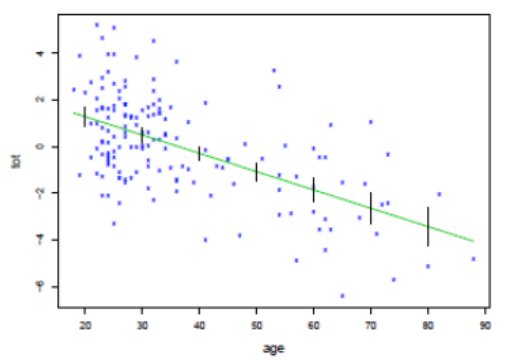
Basic setup

Problem faced: Estimating a continuous response given several explanatory variables with postulated linear relation

An example: *Consider kidney function. Data represent (x=age, y=tot, a composite measure of overall function). Kidney function declines with age and need to provide additional information concerning decline rate. Important in managing kidney transplants*

Numerical example in Lab. Here, concepts and methods.

Check: `https://en.wikipedia.org/wiki/Linear_regression`



Basic setup

Probabilistic setup

- n observations iid with K explanatory variables, one continuous response variable.
- $\mathbf{X}$ compiles n explanatory measurements (design matrix).
- $\mathbf{y}$ compiles responses.
- Exploratory analysis suggests linear relations (possibly after transformations).
- Parameters of interest:
  - $\boldsymbol{\beta} \in \mathbb{R}^k$, regression weights,
  - σ^2, noise variance

The postulated linear regression model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

$$\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n).$$

Basic setup

Classical setup

- Given $\mathbf{X}$, the likelihood of $\mathbf{y}$ is:

$$p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}, \sigma^2) = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right).$$

- The MLEs are:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}.$$

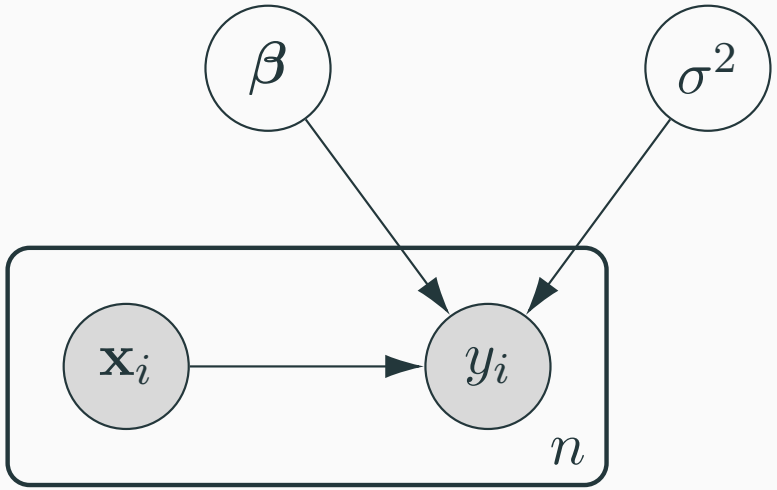
$$\hat{\sigma}^2 = \frac{1}{n}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})^\top(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}).$$

- An unbiased estimator is:

$$\hat{\sigma}^2_{\text{unbiased}} = \frac{1}{n-k}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})^\top(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}).$$

Basic setup

Probabilistic setup



Basic setup

Bayesian setup. Another source of uncertainty (or info)

- Parameters viewed as uncertain quantities
- First, use the standard non-informative (improper) prior:

$$p(\boldsymbol{\beta}, \sigma^2) \propto \frac{1}{\sigma^2}.$$

Need to aggregate both sources of information: Bayes' formula. From prior to posterior. The joint posterior is

$$p(\boldsymbol{\beta}, \sigma^2 \mid \mathbf{y}, \mathbf{X}) \propto (\sigma^2)^{-\frac{n}{2}-1} \exp\left(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\right).$$

The posterior summarizes all information available about parameters.

Basic setup

The conditional marginal posteriors

$$\boldsymbol{\beta} \mid \sigma^2, \mathbf{y}, \mathbf{X} \sim \mathcal{N}\left(\hat{\boldsymbol{\beta}},\ \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\right),$$

where $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$.

$$\sigma^2 \mid \mathbf{y}, \mathbf{X} \sim \text{Inverse-Gamma}\left(\frac{n-k}{2},\ \frac{1}{2}(\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\beta}})^\top(\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\beta}})\right). \tag{1}$$

Basic setup

Sampling from the posterior

- Compute $V_\beta = (X^T X)^{-1}, \hat{\beta}, s^2$
- Sample σ^2
- Sample $\beta|\sigma^2$

Basic uses. Inference and prediction

Basic uses

Posterior summarises all info available to perform

- Point estimation
- Hypothesis testing for coefficients
- Prediction
- Model checking
- Decisions

Focus on simulation

Basic uses. Point estimation

Posterior mean of $\boldsymbol{\beta}$

$$\mathbb{E}[\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}] = \hat{\boldsymbol{\beta}}.$$

Posterior mean of σ^2

$$\mathbb{E}[\sigma^2 \mid \mathbf{y}, \mathbf{X}] = \frac{1}{n-k-2}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})^\top(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}), \quad (n-k > 2).$$

Basic uses. Hypothesis testing for coefficients

Test for $\beta^j = 0$

- For i = 1...1000
  - Generate $\sigma_i^2 \sim \sigma^2$
  - Generate $\beta_i \sim \beta|\sigma_i^2$
- $srt\beta^j \leftarrow sort(\beta_i^j)$
- $0 \in [srt\beta_{50}^j, srt\beta_{950}^j]$?

Basic uses. Forecasting

Given a new input $\mathbf{x}_* \in \mathbb{R}^k$

Conditioning on σ^2, the predictive distribution is

$$y_* \mid \mathbf{x}_*, \mathbf{y}, \mathbf{X}, \sigma^2 \sim \mathcal{N}\left(\mathbf{x}_*^\top \hat{\boldsymbol{\beta}},\ \sigma^2\left(1 + \mathbf{x}_*^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_*\right)\right).$$

Integrating out σ^2, the predictive distribution becomes a Student-t:

$$y_* \mid \mathbf{x}_*, \mathbf{y}, \mathbf{X} \sim$$
$$t_{n-k}\left(\mathbf{x}_*^\top \hat{\boldsymbol{\beta}},\ \frac{1}{n-k}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})\left(1 + \mathbf{x}_*^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_*\right)\right).$$

Basic uses. Forecasting

Sampling from the predictive

- Sample σ^2
- Sample $\beta|\sigma^2$
- Sample $y_* \sim y|x_*, \beta, \sigma^2$

Point forecasts, Interval forecasts,...

Basic uses. Model checking

Examining plots of residuals againts explanatory variables,. . . interpretable as posterior predictive checks

Advantage: Using simulation, compute the posterior predictive distribution for any data summary

Examples in Labs

Basic uses. Decision models

Some examples of decision uses of previous models:

- Regulatory compliance: y refers to concentration of a pollutant difficult to assess; a regression model reliably predicts it from x used by an agency to impose some penalty, there being an upper bound y_0 that should not be overcome;
- candidate selection: the regressor uses y to select the m highest ranked candidates; the x variables serve as predictors with y aggregating the x through a multi-objective weighted additive value model;
- resource allocation: the regressor uses the whole predictive distribution to make an inflation prediction to, say, decide a budget allocation based on the maximum predictive expected utility principle.

The case with informative (conjugate) prior

Prior and posterior

Prior (Normal–Inverse-Gamma)

$$\begin{aligned}\boldsymbol{\beta} \mid \sigma^2 &\sim \mathcal{N}(\boldsymbol{\beta}_0, \sigma^2 \mathbf{V}_0),\\ \sigma^2 &\sim \text{Inverse-Gamma}(a_0, b_0).\end{aligned}$$

Posterior (also Normal–Inverse-Gamma)

$$\begin{aligned}\boldsymbol{\beta} \mid \sigma^2, \mathbf{y} &\sim \mathcal{N}(\boldsymbol{\beta}_n, \sigma^2 \mathbf{V}_n),\\ \sigma^2 \mid \mathbf{y} &\sim \text{Inverse-Gamma}(a_n, b_n),\end{aligned}$$

with updated parameters

$$\begin{aligned}\mathbf{V}_n &= (\mathbf{V}_0^{-1} + \mathbf{X}^\top \mathbf{X})^{-1},\\ \boldsymbol{\beta}_n &= \mathbf{V}_n(\mathbf{V}_0^{-1}\boldsymbol{\beta}_0 + \mathbf{X}^\top \mathbf{y}),\\ a_n &= a_0 + \frac{n}{2}, b_n = b_0 + \frac{1}{2}\left(\mathbf{y}^\top \mathbf{y} + \boldsymbol{\beta}_0^\top \mathbf{V}_0^{-1}\boldsymbol{\beta}_0 - \boldsymbol{\beta}_n^\top \mathbf{V}_n^{-1}\boldsymbol{\beta}_n\right).\end{aligned}$$

Predictive

For a new input $\mathbf{x}_*$:

$$y_* \mid \mathbf{x}_*, \mathbf{y} \sim t_{2a_n}\left(\mathbf{x}_*^\top \boldsymbol{\beta}_n,\ \frac{b_n}{a_n}\left(1 + \mathbf{x}_*^\top \mathbf{V}_n \mathbf{x}_*\right)\right).$$

Hierarchical models

Basic setup

Before the β's were shared. Now each observation has its β coming from a same distribution

$$\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2 \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n),$$
$$\boldsymbol{\beta} \mid \sigma^2, \tau^2 \sim \mathcal{N}(\mathbf{0}, \sigma^2 \tau^2 \mathbf{I}_k).$$

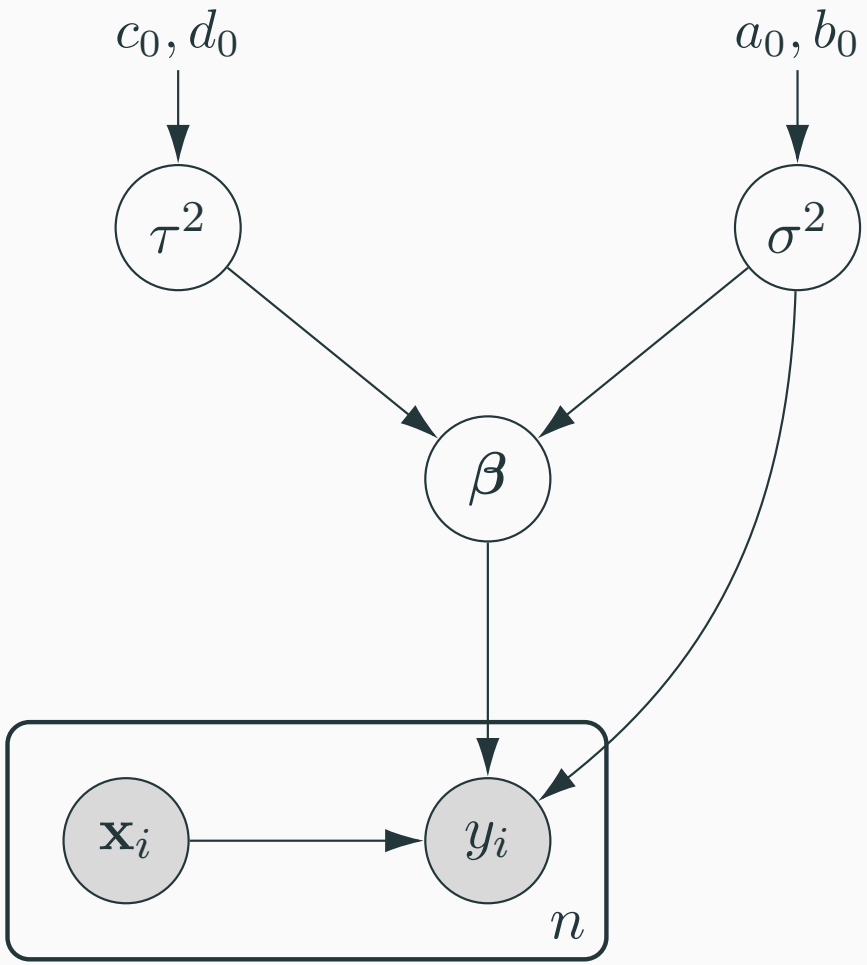
With hyperpriors

$$\sigma^2 \sim \text{Inverse-Gamma}(a_0, b_0),$$
$$\tau^2 \sim \text{Inverse-Gamma}(c_0, d_0).$$

The joint posterior is

$$p(\boldsymbol{\beta}, \sigma^2, \tau^2 \mid \mathbf{y}) \propto p(\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2)\, p(\boldsymbol{\beta} \mid \sigma^2, \tau^2)\, p(\sigma^2)\, p(\tau^2).$$

Basic setup



Key comments

- This hierarchical model induces shrinkage on β
- Closed-form joint posterior is not available, but conditionals are available enabling MCMC analysis (Chapter 6 and Labs 6).

Empirical Bayes Methods

Empirical Bayes methods

In the previous hierarchical model, Empirical Bayes replaces the hyperprior on τ^2 by a *point estimate*, obtained from the data typically by maximizing the marginal likelihood:

$$p(\mathbf{y} \mid \tau^2, \sigma^2) = \int p(\mathbf{y} \mid \boldsymbol{\beta}, \sigma^2)\, p(\boldsymbol{\beta} \mid \sigma^2, \tau^2)\, d\boldsymbol{\beta}.$$

yielding $\mathbf{y} \mid \tau^2, \sigma^2 \sim \mathcal{N}\left(\mathbf{0},\ \sigma^2(\mathbf{I}_n + \tau^2 \mathbf{X}\mathbf{X}^\top)\right)$, and $\hat{\tau}^2 = \arg\max_{\tau^2}\ p(\mathbf{y} \mid \tau^2, \sigma^2)$. Plugging $\hat{\tau}^2$ into the prior,

$$\boldsymbol{\beta} \mid \sigma^2, \mathbf{y}, \hat{\tau}^2 \sim \mathcal{N}(\boldsymbol{\beta}_{EB}, \sigma^2 \mathbf{V}_{EB}),$$

where $\mathbf{V}_{EB} = (\mathbf{X}^\top \mathbf{X} + \hat{\tau}^{-2} \mathbf{I}_k)^{-1}$, $\boldsymbol{\beta}_{EB} = \mathbf{V}_{EB} \mathbf{X}^\top \mathbf{y}$

Key comments

- Empirical Bayes estimates hyperparameters (here τ^2) from the data.
- The resulting estimator $\boldsymbol{\beta}_{EB}$ is a shrinkage estimator.
- Unlike full Bayesian inference, uncertainty in τ^2 is not propagated.
- If σ^2 is also unknown, one may estimate (τ^2, σ^2) jointly.

Final comments

Final comments

- Again we have used conjugate priors. But in several cases we were not able to derive the posteriors.... We see later on how to deal with them computationally
- Important machine learning models like LASSO or ridge regression may be seen as MAPs in Bayesian LR with certain priors.
- Linear regression models constitute the simplest example of supervised learning models for regression.
- Many NN models are non-linear regression models.

References

References

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Lab 4

Lab4

Linear regression in R